

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Analysis of Training and Testing MSE

Given:

- **Training Mean Squared Error (MSE) = 0.85**
- **Testing Mean Squared Error (MSE) = 5.20**

The testing error is much higher than the training error, indicating that the neural network performs well on the training data but poorly on unseen test data.

Possible Reasons

1. Overfitting (Most Likely Cause)

- The model has learned the training data too closely, including its noise and outliers.
- It memorizes the training samples instead of learning general patterns.
- As a result, it performs well on training data but poorly on new data.

2. Complex Model

- The neural network may have too many hidden layers or neurons.
 - A highly complex model has a large number of parameters, making it prone to overfitting.
 - **3. Insufficient Training Data**
 - If the training dataset is too small, the model cannot learn the true distribution of the data.
 - It may fit the limited samples rather than generalizing.
 - **4. Noise or Outliers**
 - The training data may contain incorrect labels or extreme values.
 - The model may learn these noisy patterns, reducing performance on test data.
 - **5. Data Distribution Mismatch**
 - The training and testing datasets may come from different distributions.
 - For example, the training data may represent one group of students while the testing data represents another with different characteristics.
 - **6. Lack of Regularization**
 - Techniques such as Dropout, L1/L2 regularization, or Early Stopping may not have been used.
 - Without regularization, the model easily overfits.
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Corrective Measures

1. Apply Regularization

- Use **L1 or L2 regularization** to penalize large weights.
- This reduces model complexity and improves generalization.

2. Use Dropout

- Randomly deactivate a percentage of neurons during training.
- Prevents neurons from becoming overly dependent on each other and reduces overfitting.

3. Early Stopping

- Monitor the validation loss during training.
- Stop training when the validation error starts increasing, preventing overfitting.

4. Increase Training Data

- Collect more training samples or use **data augmentation** where applicable.
- More data helps the model learn general patterns.

5. Reduce Model Complexity

- Decrease the number of hidden layers or neurons.
- A simpler model is less likely to memorize the training data.

6. Cross-Validation

- Use **k-fold cross-validation** to evaluate the model on multiple data splits.
- This provides a more reliable estimate of model performance.

7. Feature Selection and Data Cleaning

- Remove irrelevant features, noise, and outliers.
- Normalize or standardize input features to improve learning.

8. Hyperparameter Tuning

- Adjust the learning rate, batch size, number of epochs, optimizer, and network architecture to find the best-performing model.

Conclusion

The large difference between the **training MSE (0.85)** and **testing MSE (5.20)** strongly indicates **overfitting**. The model has learned the training data very well but fails to generalize to unseen data. This can be improved by applying **regularization techniques (Dropout, L1/L2), Early Stopping, collecting more training data, simplifying the network architecture, cleaning the data, and performing proper hyperparameter tuning.**

Final Answer:

- **Training MSE = 0.85 (low)** → Good performance on training data.
- **Testing MSE = 5.20 (high)** → Poor generalization due to **overfitting**.
- **Recommended corrective measures:** Dropout, L1/L2 regularization, Early Stopping, increase training data, reduce model complexity, perform cross-validation, clean the dataset, and tune hyperparameters.

Code of Conduct:

I SHAM S. (192325076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (23)	Level 1 – Beginning (01)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors

Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided